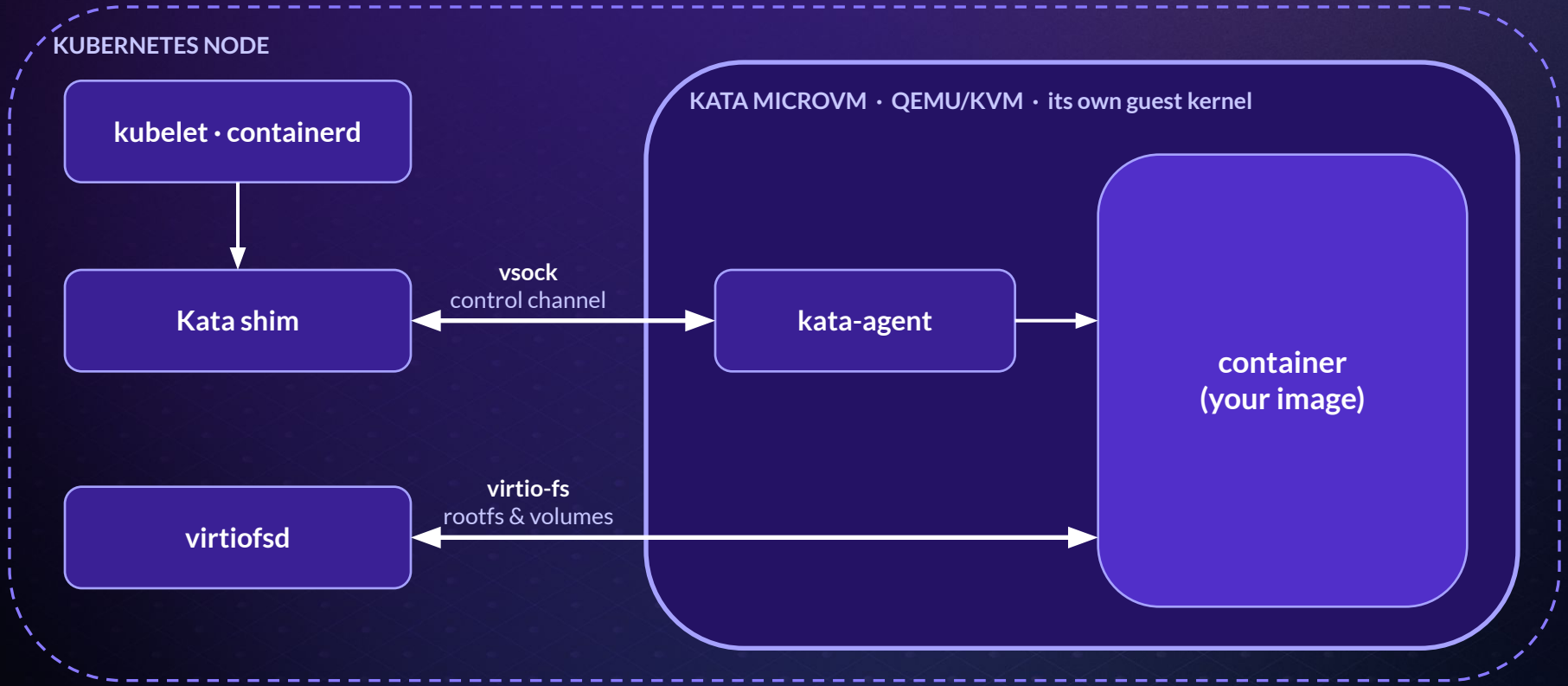
Teleport | The AI Infrastructure Identity Company

Building an Identity Runtime with Teleport and Kata Containers

Boston Kubernetes Meetup · October 2026

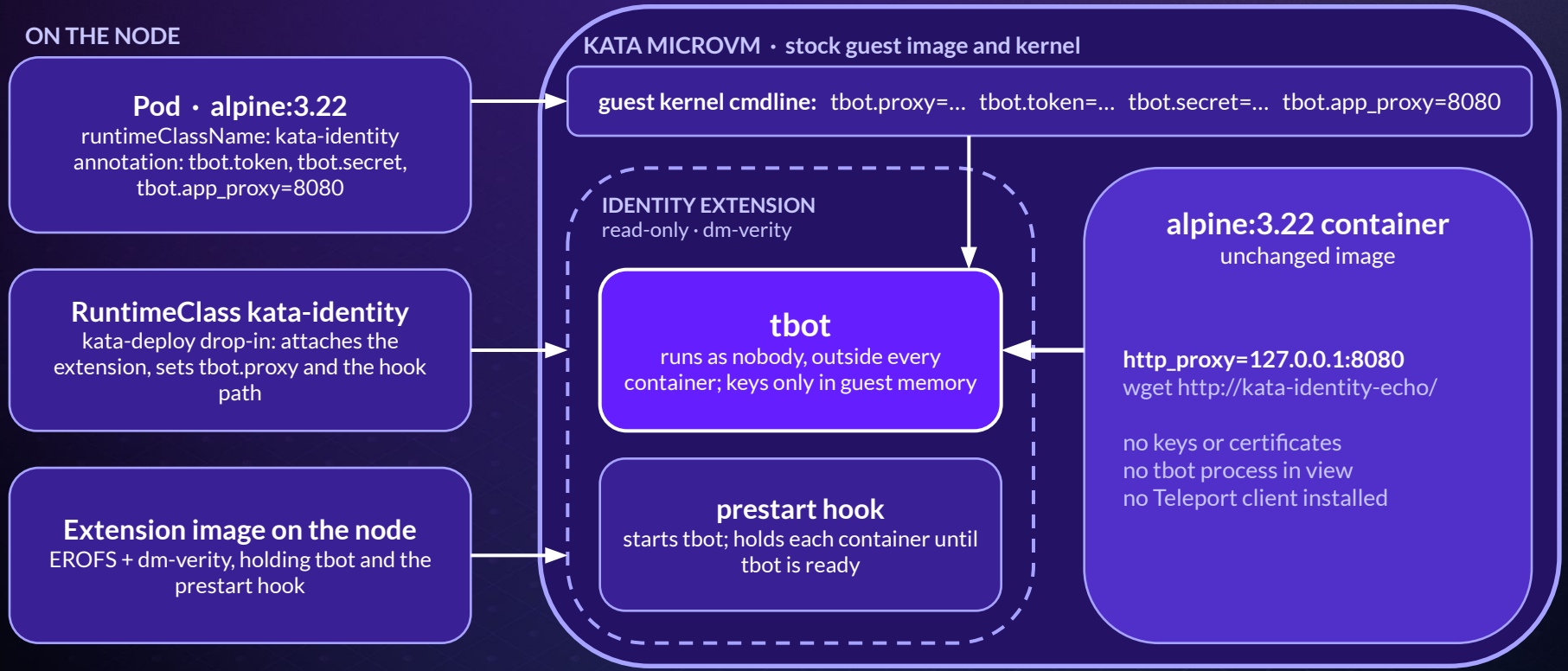
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How Kata Containers run a pod



Teleport

kata-identity: where the identity lives

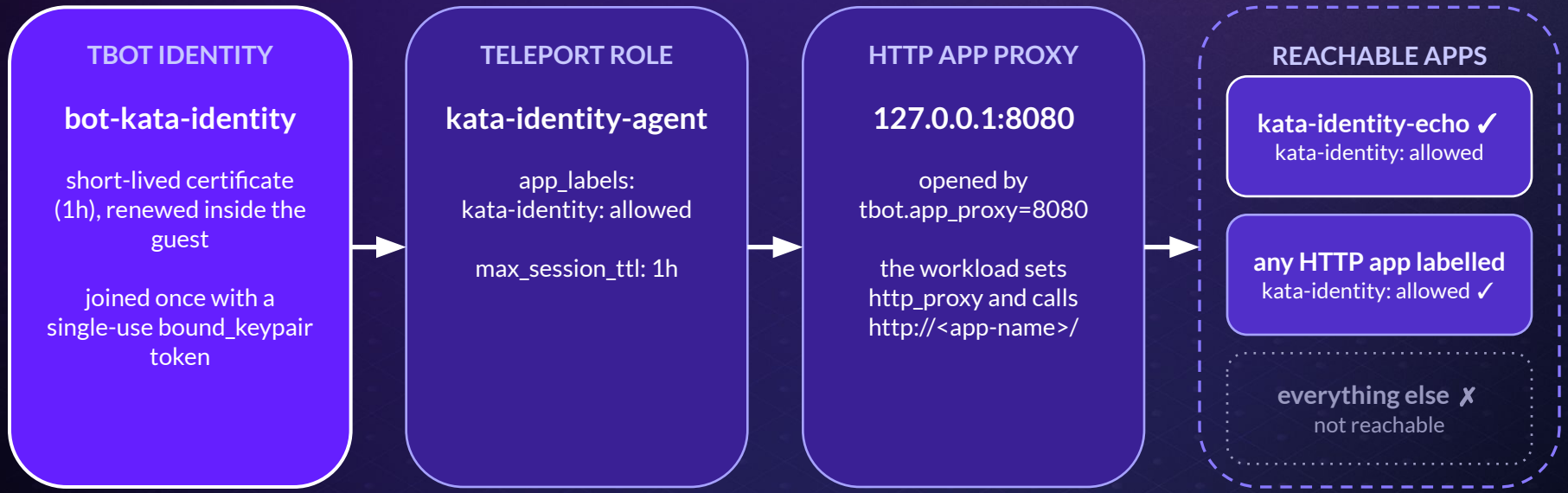


Teleport

Why this is cool

- **Not in the Kubernetes cluster.** No Secret, no ServiceAccount token, no sidecar. tbot runs in the guest, outside every container, and its keys live only in guest memory.
- **Locked identity.** A single-use bound-keypair token is spent before the container starts. tbot ships in a dm-verity extension and runs as nobody; the workload never sees a key.
- **Network policy does the rest.** Allow egress only to the Teleport proxy and Teleport becomes the only way out, enforced on the host by the CNI.
- **The Teleport role sets the reach.** One app-proxy listener reaches every Teleport-protected HTTP app the role's labels allow. Change the role, not the pod.

From tbot identity to reachable apps



Reach is granted in Teleport, not in the pod: change the role and every pod on the runtime follows.
Each request is made as bot-kata-identity and audited. The echo app shows the signed JWT: sub=bot-kata-identity.

Teleport

NetworkPolicy: Teleport is the only way out

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: teleport-only
  namespace: agents
spec:
  podSelector: {}            # every pod in the namespace
  policyTypes: [Ingress, Egress]   # no ingress rules: nothing
in
  egress:
  - to:                      # DNS, to resolve the proxy
    - namespaceSelector: {}
      podSelector:
        matchLabels: {k8s-app: kube-dns}
    ports:
    - {port: 53, protocol: UDP}
    - {port: 53, protocol: TCP}
  - to:                      # the Teleport proxy, nothing
else
    - ipBlock: {cidr: 203.0.113.10/32}
    ports:
    - {port: 443, protocol: TCP}
```

- **Covers tbot too.** tbot and the workload share the pod's network namespace, so one policy governs both.
- **The guest can't turn it off.** The CNI enforces it on the host, outside the microVM.
- **The workload can reach the proxy, but holds no credentials.** Its only working path is tbot's listener.
- **Proxy IPs change (Teleport Cloud)** Use Cilium's toFQDNs instead of an ipBlock.

Thank you!



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Booth 103

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Platform Engineering Day

Find us at **Table 7**

MON · NOV 9

KubeAuto Day

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Teleport

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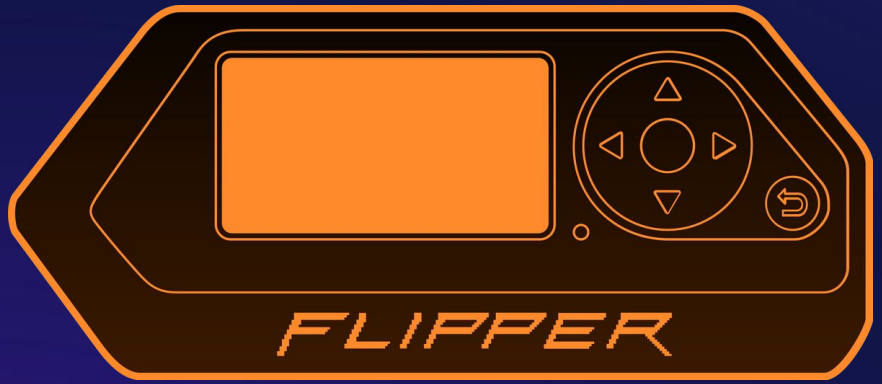
Teleport

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